

Our problem statement:

Are meteorological, agricultural, and satellite features (i.e. from the CY-Bench Dataset) good predictors of European Wheat Futures Price Changes?

Research Report Structure:

Advice from the formative and summative project briefs:

- The report should be structured in the way that best fits the story we're telling
 - o Intro (background info on the dataset & prediction problem at hand) → define context and scope
 - o Brief exploratory data analysis → give information you took out from exploring the dataset's data that is important in you pursuing the task (or is of explicit importance)
 - o Information about all the aspects of our analysis (Bulk of the report): data preprocessing → quantitative results → model comparison
 - o Conclusion discussing: Learnings, ideas for further improvement, etc.
- Literature review can be included if relevant to our Dataset

Past MA429 Projects:

- <https://github.com/ZXChen299/Previous-LSE-MA429-Projects/blob/master/MA429%20Final%20Project/IMDB/Report.pdf>
 - o **Note: the report is overall not great**
 - o Structure of main body: Intro (subsections: Motivation, Goal and Limitations, Structure of the report), Data Pre-processing (subsections: preparation of the dataset, exploratory analysis on the training set), Performance Measures (subsections: regression, classification), Regression (subsections: simple and multiple regression, estimating and interpreting model coefficients, issues and improvements to the regression model, making predictions from the linear model), Classification (subsections: model tuning and training, application and performance on testing dataset), Conclusion (subsections: interpretation of results)
- No others were found

From research:

- Advice from Carnegie Mellon: <https://www.stat.cmu.edu/~brian/701/notes/paper-structure.pdf>
 - o Don't use complex language
 - o Appendix = more detailed and process-oriented
 - o Proposed structure:
 - **Intro containing** = summary of the study and data + any relevant substantive context / background / framing issues
 - The Big Questions answered by our data analysis and a summary of our conclusions about these questions
 - Brief outline of the remainder of the paper
 - Body = can be traditional (data/methods/analysis/results) or question oriented (i.e. one section per question raised in the introduction)

General Research Report Advice:

- 16 pages limit (excluding title page, executive summary, table of contents, bibliography and appendices)
- No appendix page limit but only include relevant things / things of interest

Proposed Main Body Structure

- **Introduction** = data set key info, chosen prediction task and justification, context and scope of the task, summary of the study conducted and results (1 page)
- **(Optional) Brief Literature Review** = review some of the key papers on the dataset as well as the prediction task at hand (could help set expectations for the reader) – (1 page)
- **Data Pre-processing** = Details of how we managed the regional data split when fitting our models (maybe: method 1 = fit one model per region, method 2 = aggregation) (2 pages)
- **Redefining the prediction task** = important section to explain how we change the prediction task → here we're addressing questions such as: why are we predicting 5 day ahead returns and not daily returns? How are we dealing with the futures rolling cost in our time series price? → this should be brief and clear (1 page)
- **Feature engineering and selection** = How we augmented our data to add predictive power, using insights from an agricultural (e.g. if temperature is consistently high for a period and precipitation is low, this may indicate a drought) | also we should indicate how we performed feature selection before fitting all of our models (e.g. using a decision tree and looking at feature importance / e.g. putting features on the same scale and looking at their correlation) (2 pages)
- **For each method used** =
 - o Method justification = why we believe this method would improve predictive power
 - o Analysis = how we tuned the method, our methodology and justification
 - o Results = what was the predictive power of this method(4 times 2 pages)
- **Model Results Comparison** = Comparing and interpreting results across models, proposing explanations for outcomes (1 page)
- **Conclusion** = Discussing results, drawing some key learnings from the discussion, ideas for further improvement, proposed future work (1 page)